Series solutions of higher order equations

When we want to find the series solution to a higher order differential equation, we'll of course need the series representations of higher order derivatives.

We saw previously the power series representations in x of y, y', and y'', so let's add just a couple of derivatives to that list.

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots$$

$$y''' = \sum_{n=3}^{\infty} c_n n(n-1)(n-2) x^{n-3} = 6c_3 + 24c_4 x + 60c_5 x^2 + 120c_6 x^3 + 210c_7 x^4 + \dots$$

$$y^{(4)} = \sum_{n=4}^{\infty} c_n n(n-1)(n-2)(n-3) x^{n-4} = 24c_4 + 120c_5 x + 360c_6 x^2 + 840c_7 x^3 + \dots$$

Just as we did before, we can use these formulas to find the power series solution to a higher order differential equation.

Keep in mind that, the higher the order of the differential equation, the less likely it becomes that we'll be able to find clean series representations when we start plugging values from the index into the recurrence relation.

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That being said, let's do an example so see what kind of a power series solution we can find for a higher order differential equation.

Example

Find a power series solution in x to the differential equation.

$$y''' + y' - xy = 0$$

We'll substitute y, y', and y''' into the differential equation.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^{n-1} - x\sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Let's check to see if the series are in phase. Because the index of the first series starts at $n = 3$, the first series begins with an x^0 term. Because the index of the second series starts at $n = 1$, the second series begins with an x^0 term. And because the index of the third series starts at $n = 0$, the third series begins with an x^1 term.

So to put these series in phase, we'll pull the x^0 term out of both the first and second series.

$$c_3 3(3-1)(3-2)x^{3-3} + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3}$$

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$$+c_1 1x^{1-1} + \sum_{n=2}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

$$c_1 + 6c_3 + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=2}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, $k = n - 1$ and $n = k + 1$ into the second series, and $k = n + 1$ and $n = k - 1$ into the third series.

$$c_1 + 6c_3 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+3-1)(k+3-2)x^k$$

$$+\sum_{k=1}^{\infty} c_{k+1}(k+1)x^k - \sum_{k=1}^{\infty} c_{k-1}x^k = 0$$

$$c_1 + 6c_3 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + \sum_{k=1}^{\infty} c_{k+1}(k+1)x^k - \sum_{k=1}^{\infty} c_{k-1}x^k = 0$$

With the series in phase and matching indices, we can finally add them.

$$c_1 + 6c_3 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + c_{k+1}(k+1)x^k - c_{k-1}x^k = 0$$

$$c_1 + 6c_3 + \sum_{k=1}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_{k+1}(k+1) - c_{k-1}]x^k = 0$$

The $c_1 + 6c_3$ value in front of the series is associated with the $k = 0$ term, while all of the other $k = 1, 2, 3, \ldots$ terms are still in the series.

$$k = 0 \qquad c_1 + 6c_3 = 0$$

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$$c_1 = -6c_3 \text{ and } c_3 = -\frac{c_1}{6}$$

$k = 1, 2, 3, \ldots$ $\qquad c_{k+3}(k+3)(k+2)(k+1) + c_{k+1}(k+1) - c_{k-1} = 0$

We'll solve the recurrence relation for the coefficient with the largest subscript. This recurrence relation includes c_{k+3}, c_{k+1}, and c_{k-1}, so we'll solve for c_{k+3}.

$$c_{k+3} = \frac{c_{k-1} - c_{k+1}(k+1)}{(k+3)(k+2)(k+1)}$$

Now we'll start plugging in values $k = 1, 2, 3, \ldots$.

$$k = 0 \qquad c_3 = -\frac{c_1}{3!}$$

$$k = 1 \qquad c_4 = \frac{c_0 - 2c_2}{4!} \qquad\qquad k = 2 \qquad c_5 = \frac{3c_1}{5!}$$

$$k = 3 \qquad c_6 = -\frac{c_0 - 8c_2}{6!} \qquad\qquad k = 4 \qquad c_7 = -\frac{7c_1}{7!}$$

$$k = 5 \qquad c_8 = \frac{6c_0 - 18c_2}{8!} \qquad\qquad k = 6 \qquad c_9 = \frac{25c_1}{9!}$$

$$\vdots \qquad\qquad\qquad\qquad \vdots$$

There's not an obvious pattern here. As we mentioned earlier, the higher the order of the differential equation, the less likely it becomes that we'll be able to find clean series representations.

So without worrying about the pattern, we'll just use these values to find that the general solution is

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$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + \dots$$

$$y = c_0 + c_1x + c_2x^2 - \frac{c_1}{3!}x^3 + \frac{c_0 - 2c_2}{4!}x^4 + \frac{3c_1}{5!}x^5 - \frac{c_0 - 8c_2}{6!}x^6 + \dots$$

$$y = c_0\left(1 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right) + c_1\left(x - \frac{1}{3!}x^3 + \frac{3}{5!}x^5 + \dots\right)$$

$$+c_2\left(x^2 - \frac{2}{4!}x^4 + \frac{8}{6!}x^6 + \dots\right)$$

$$y = c_0\left(1 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right) + c_1x\left(1 - \frac{1}{3!}x^2 + \frac{3}{5!}x^4 + \dots\right)$$

$$+c_2x^2\left(1 - \frac{2}{4!}x^2 + \frac{8}{6!}x^4 + \dots\right)$$

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